

SCOTECI: Synthetic Chain-of-Thought Demonstrations for Document-Level Event Causality Identification

[TODO: working title — alternatives: “Documents, not Pairs: ...” / “Rationalized Few-Shot Prompting for Document-Level ECI”]

Abstract

- LLMs applied to Event Causality Identification (ECI) suffer from causal hallucination: near-total recall, unusable precision [Gao et al., 2023].
- Prior LLM scaffolds (Dr.ECI, SERE) fight this at the *pair* level — one call per event pair, bare input→label demonstrations.
- We propose SCOTECI: (i) reformulate LLM-based ECI at the *document* level — one call labels every pair of a document jointly under a structured reasoning protocol with explicit coherence rules; (ii) demonstrations are whole neighbour documents equipped with *synthetic chain-of-thought*: gold-guided reasoning generated by an LLM, then decontaminated so it reads as genuine discovery.
- No fine-tuning, no human rationales, no external knowledge base.
- Headline: on Causal-TimeBank, F1 51.0 vs. 20.0 for the best prior LLM method, with balanced precision/recall instead of the degenerate high-recall regime; best precision on MAVEN-ERE.

1 Introduction

- **Task and stakes.** ECI = given a document and event mentions, decide for event pairs whether a (directed) causal relation holds [Cheng et al., 2025]. Feeds event forecasting, QA, narrative understanding.
- **Two paradigms, two failure modes.**
 - Fine-tuned encoders (BERT/RoBERTa-era, graph- and knowledge-enhanced) are strong in-domain but need per-corpus annotation and generalize poorly across domains [Cai et al., 2025, Hao et al., 2026].
 - Prompted LLMs need no training but exhibit *causal hallucination* [Gao et al., 2023]: on Causal-TimeBank, recall 80–96% with precision below 10% (Table 1, Base/CoT rows).
- **Prior LLM scaffolds are pair-level.** Dr.ECI decomposes each pair into a four-agent discovery/reasoning pipeline [Cai et al., 2025]; SERE retrieves structurally similar *pair* examples via ConceptNet paths, syntax-tree edit distance and causal-pattern filtering [Hao et al., 2026]; DICP learns deep in-context prompts per pair [Mu et al., 2025]. Consequences:
 - $O(|\text{pairs}|)$ LLM calls per document, each re-paying the full context ($O(n^2)$ in the number of mentions).

- Each pair judged in isolation: no symmetry/transitivity consistency, no confounder check against the other pairs, the document’s causal graph is never seen as a whole.
- The scaffolds were engineered around the weaknesses of 2023–2024-era models; as base models improve, the scaffold becomes the ceiling, not the fix.
- **Idea 1 — operate at the document level.** One call labels *all* pairs of a document jointly, under an in-prompt protocol (mention pre-filter → per-pair for/against arguments → plausibility ranking → final JSON) plus explicit coherence rules (symmetry, dataset-specific transitivity). This shifts the burden from external orchestration to the model’s own long-context reasoning — deliberately betting on model quality, so the method *improves as models improve*.
- **Idea 2 — synthetic chain-of-thought few-shots.** Document-level demonstrations are whole labelled neighbour documents (TF-IDF-retrieved from train). Bare input→gold-JSON demos teach the output format, not the reasoning; human-written document-level rationales do not exist. We therefore *generate* them:
 - gold-guided generation: the demo document is presented under the exact task prompt together with its gold labels; the LLM writes reasoning that leads naturally to those labels;
 - decontamination: the reasoning is rewritten from scratch to purge any foreknowledge phrasing, so every sentence reads as discovery, not confirmation;
 - optional 3-step variant: blind attempt → gold-based correction → decontamination (preserves the texture of a genuine attempt);
 - injected as multi-turn (user, assistant) exchanges; cached per demo document.
- **Results preview.** Single evaluation table (Table 1): CTB F1 51.0 vs. 20.0 (SERE) — +31 F1 with precision 53.2 vs. ≤ 13.8 for every LLM baseline; best MAVEN-ERE precision; competitive ESC. [TODO: reconcile ESC/MAVEN full-document (intra+inter) numbers before submission — currently intra-sentence only.]
- **Contributions.**
 - A document-level reformulation of LLM-based ECI with a structured reasoning protocol and in-prompt coherence rules — $O(1)$ calls per document, globally consistent predictions.
 - A synthetic-CoT demonstration pipeline (gold-guided generation + decontamination), model-agnostic, requiring no human rationales and no knowledge base.
 - State-of-the-art LLM results on CTB by a wide margin, balanced precision/recall across datasets, plus ablations isolating the contribution of each component and a cost analysis.

2 Related Work

2.1 Supervised ECI

- Sentence-level fine-tuned models, often knowledge- or structure-enhanced: prompt-tuning with constrained attention [Zhang et al., 2023a], deterministic-assumption prompt learning [Xiang et al., 2023], LLM-derived knowledge + concept-level relations [Su et al., 2025],

knowledge-interaction graphs [Hu et al., 2025], contrastive representation learning [Liao et al., 2026], information-bottleneck refinement [Zhao et al., 2026].

- Document-level ECI is well established *in the supervised world*: event relational graph transformers [Chen et al., 2022], sparse event representations for discriminative reasoning [Yuan et al., 2023], iterative identify-while-learn frameworks [Liu et al., 2024], zero-shot cross-lingual transfer [He et al., 2024], knowledge-guided binary QA [Wang et al., 2024b].
- Gap: strong in-domain, but require per-corpus training data; and the LLM-prompting line of work *regressed* to pair-level inference — we bring the document-level formulation to prompting.

2.2 LLM-based ECI

- Gao et al. [2023]: systematic evaluation; ChatGPT-class models over-predict causality (causal hallucination); established the preprocessing and evaluation protocol we and our baselines follow.
- Dr.ECI [Cai et al., 2025]: multi-agent decomposition per pair — discovery stage (Causal Explorer, Mediator Detector) and reasoning stage (Direct/Indirect Reasoner) guided by causal-pattern DAG priors.
- SERE [Hao et al., 2026]: structural example retrieval per pair — ConceptNet shortest-path edit distance + dependency-tree edit distance + causal-pattern filtering; balanced positive/negative demonstrations; demonstrations are bare input→label pairs.
- Synthetic control for ECI [Wang et al., 2024a]: counterfactual estimation flavour; DICP [Mu et al., 2025]: learned deep in-context prompts.
- Positioning: all of the above (i) prompt per pair, (ii) show label-only demonstrations. SCOTECI prompts per document and shows *reasoning-bearing* demonstrations.

2.3 In-Context Learning and Synthetic Rationales

- CoT prompting boosts multi-step reasoning [Wei et al., 2022]; ICL is highly sensitive to demonstration choice and content [Liu et al., 2022, Min et al., 2022].
- Auto-CoT generates rationales with zero-shot-CoT over clustered queries [Zhang et al., 2023b]; STaR bootstraps rationales via *rationalization* — generating reasoning conditioned on the known answer — but consumes them as fine-tuning data [Zelikman et al., 2022]; step-by-step distillation extracts rationales as training signal [Hsieh et al., 2023].
- Ours differs: rationalization used *at inference time* as ICL demonstrations, at document scale, with an explicit decontamination pass to remove answer-leakage phrasing — without which the demonstrations teach the model to assert rather than reason.

3 Task Formulation

- Input: document D with inline-marked event mentions $M = \{m_1, \dots, m_n\}$ (<ID word> format), and a pair set P : the annotated candidate list when the corpus provides one (constrained setting), otherwise all unordered mention pairs (open-ended setting).

- Output: a complete labelling $\ell : P \rightarrow \{\textit{causes}, \textit{causedby}, \textit{norel}\}$ — a unified directed label space into which each corpus’ native labels are mapped [TODO: spell out per-dataset label mapping (CAUSE/PRECONDITION, FALLING_ACTION, CauseEffect/EffectCause)].
- Document-level inference: the model produces ℓ for *all* of P in a single generation, as a JSON array; contrast with pair-level inference which issues $|P|$ independent queries.
- Evaluation: micro precision/recall/F1 over the positive (causal) classes; *norel* excluded; intra- vs. inter-sentence subsets where the corpus distinguishes them; preprocessing follows Gao et al. [2023] for comparability with Hao et al. [2026] and Cai et al. [2025].

4 Method

4.1 Overview

- Pipeline (Figure ??): test document $\rightarrow$ retrieve k neighbour training documents $\rightarrow$ generate (and cache) synthetic CoT for each neighbour $\rightarrow$ assemble multi-turn prompt (demonstrations as past exchanges, then the test document) $\rightarrow$ single document-level LLM call $\rightarrow$ parse JSON $\rightarrow$ optional resampling with majority vote.
- Two components carry the contribution: the document-level structured prompt (§4.2) and the synthetic-CoT demonstrations (§4.4); retrieval (§4.3) and aggregation (§4.5) are deliberately simple.

4.2 Document-Level Structured Prompting

- One call per document; every pair of P requested at once; the model sees the full narrative, not a windowed snippet.
- In-prompt reasoning protocol (four ordered steps, enforced by the system prompt):
 - Step 1 — mention-level pre-filter: affirmatively argued exclusions (“ m is a location with no agentive role”), never blanket dismissal; surviving mentions listed.
 - Step 2 — per-pair arguments *for* (connective, mechanism, counterfactual asymmetry, discourse structure) and *against* (temporal-only, confounder, reporting verb, distance).
 - Step 3 — qualitative ranking of retained pairs by causal plausibility, no labels committed yet.
 - Step 4 — final JSON over *every* requested pair; everything pre-filtered defaults to *norel*.
- Precision-first rule block (targets causal hallucination directly): two-signal rule for non-overt links; stricter gate for inter-sentential / long-range pairs; default-to-*norel* under uncertainty; confounder / proximal-cause check; reporting and temporal connectives explicitly ruled non-causal; counterfactual direction test.
- In-prompt coherence rules: symmetry ($\textit{causes}(i, j) \Leftrightarrow \textit{causedby}(j, i)$) and dataset-specific transitivity rules — checkable at parse time because the whole graph is produced jointly [TODO: decide whether the coherence checker/post-hoc rule pruning appears in main text or appendix].
- Why this leverages better models: the protocol asks for exactly the global, long-context reasoning that frontier models are differentially good at; pair-level scaffolds cap that ability at the width of one pair.

4.3 Neighbour-Document Retrieval

- Demonstration pool: training split of the same corpus (capped pool size to bound CoT-generation cost) [TODO: final pool size].
- Selection: TF-IDF cosine over full document text, top- k ($k = 3$); alternatives evaluated in §6.1: random, mention-set Jaccard, sentence-embedding cosine.
- Demonstrations are *whole documents with their complete gold labelling*, including the *norel* pairs — every demo carries explicit negative evidence, the pair-level analogue of which SERE must engineer via balanced retrieval.

4.4 Synthetic Chain-of-Thought Generation

- Motivation: label-only demos teach the output schema, not the reasoning; document-level human rationales are unavailable and prohibitively expensive to write; gold labels for training documents are available — so rationales can be *synthesized*.
- Default 2-step protocol, per demonstration document:
 - (i) *gold-guided generation*: the demo document is shown under the exact task prompt, plus its gold labels and the instruction to write reasoning that “leads naturally to these labels from the text alone” without revealing foreknowledge;
 - (ii) *decontamination*: the response is rewritten from scratch to purge leakage phrasing (“the labels show. . .”, “as indicated by the expected output. . .”), so each sentence reads as analysis, not confirmation.
- 3-step variant: (i) blind attempt with no gold access, (ii) gold-based correction that fixes wrong conclusions while keeping correct reasoning, (iii) decontamination — preserves the texture and difficulty profile of a genuine attempt [TODO: report 2- vs. 3-step comparison in §6.1].
- Injection: each demo becomes a (user task turn, assistant CoT turn) exchange preceding the test document — the model “remembers having done” k documents correctly, reasoning included; compatible with multi-step task prompts (one synthetic response per step).
- Generated once per (document, protocol), cached across the run; generator model = inference model by default (self-generated rationales) [TODO: teacher-vs-self generation experiment — does a stronger generator lift a weaker reasoner?].
- Relation to STaR-style rationalization [Zelikman et al., 2022]: same trick — condition on the answer to obtain reasoning — but used as in-context demonstrations rather than fine-tuning data, and with an explicit decontamination pass, which we show is load-bearing (§6.1).

4.5 Inference and Aggregation

- Temperature 0; strict JSON parsing with bounded retries; missing pairs default to *norel*.
- Optional resampling: N independent passes with per-pair majority vote, ties broken to *norel* (precision-preserving); pair order shuffled per pass to break position bias [TODO: state whether the main table uses resampling and with which N].

- Cost: 1 call per document (N with resampling) vs. $|P|$ calls for pair-level methods — plus SERE’s per-pair retrieval infrastructure (ConceptNet + parser) and Dr.ECI’s 4 agent calls per pair; quantified in §6.2.

5 Experiments

5.1 Datasets and Metrics

- EventStoryLine (ESC) [Caselli and Vossen, 2017]: 22 topics, 258 docs, 1,770 causal pairs [TODO: confirm version used (0.9 vs 1.5) and align statement with Gao preprocessing].
- Causal-TimeBank (CTB) [Mirza et al., 2014]: 183 docs, 318 causal pairs — sparsest positives, hence the dataset where hallucination is most punishing; no intra/inter split.
- MAVEN-ERE [Wang et al., 2022]: large-scale Wikipedia corpus, 57,992 causal pairs [TODO: state evaluated subset/sample size].
- Preprocessing follows Gao et al. [2023]; micro P/R/F1 on the causal classes, *noel* excluded.
- [TODO: decide whether the MECI multilingual evaluation [Lai et al., 2022] goes in (appendix?) — pipeline supports it.]

5.2 Baselines

- Base: task description only; CoT [Wei et al., 2022]: step-by-step before answering; Dr.ECI [Cai et al., 2025]; SERE [Hao et al., 2026] — all pair-level.
- Baseline scores on GPT-4o-mini and Gemini-1.5-pro reproduced from Hao et al. [2026] (their Table 2); Dr.ECI PaLM2 row from Cai et al. [2025]. We do not re-run them.
- [TODO: consider adding a same-backbone Base/CoT row (our model, pair-level) so the doc-level effect is separable from the backbone effect — currently the table conflates them.]

5.3 Implementation Details

- Backbone: [TODO: state final backbone(s) — e.g. DeepSeek-V3.2 via OpenRouter — and justify the “leverage better models” claim with at least one frontier + one mid-tier model].
- $k = 3$ demonstrations, TF-IDF selection, 2-step CoT protocol, temperature 0 [TODO: re-sampling config for final runs; pool size; exact prompt files versioned in the repo].
- Synthetic CoT generation cost: [TODO: 2 (or 3) extra calls per unique demo document, amortized by caching — give measured token counts].

5.4 Main Results

- **CTB (the headline).** F1 51.0 vs. 20.0 for the best baseline (SERE): +31 F1. Precision 53.2 where no LLM baseline exceeds 13.8 — the degenerate high-recall/low-precision regime is broken, on the dataset where positives are rarest and hallucination costliest.
- **MAVEN-ERE.** Best precision in the table (35.5); F1 36.6 trails SERE’s 42.3 but the comparison is not yet apples-to-apples (intra-sentence-only, see caveat) [TODO: full-dataset run].

Table 1: Main results. Baseline rows reproduced from Hao et al. [2026] (their Table 2). Best score per column is in bold across all rows. †: results from Cai et al. [2025]. CTB has no intra/inter split. For ESC and MAVEN-ERE, baseline scores are dataset-level, while SCOTECI (*) reports intra-sentence-only results for now – not yet directly comparable on those two columns.

Method	ESC			CTB			MAVEN-ERE		
	P	R	F1	P	R	F1	P	R	F1
PaLM2:									
Dr.ECI†	29.0	75.4	41.9	-	-	13.0	22.0	80.0	34.5
GPT-4o-mini:									
Base	28.4	82.9	42.3	5.4	80.5	10.1	23.1	91.5	36.9
CoT	29.4	80.6	43.1	6.3	79.6	11.6	25.4	87.9	39.4
Dr.ECI	31.0	90.2	46.1	8.2	91.2	15.1	26.0	93.0	40.7
SERE	48.3	51.5	49.9	13.8	36.3	20.0	34.5	54.6	42.3
Gemini-1.5-pro:									
Base	24.0	80.8	37.0	4.7	95.6	9.0	21.2	95.1	34.6
CoT	25.1	85.4	38.7	4.7	88.5	8.9	23.3	89.2	37.0
Dr.ECI	27.6	82.0	41.3	7.2	88.5	13.3	23.4	91.7	37.3
SERE	36.5	59.3	45.2	9.9	69.9	17.4	29.9	60.2	39.9
Ours:									
SCOTECI*	34.7	57.2	43.2	53.2	49.0	51.0	35.5	37.7	36.6

- **ESC.** 43.2, in the CoT-baseline range, below SERE’s 49.9 — same caveat [TODO: full-dataset run; if the gap persists, discuss: ESC’s dense per-document annotation may reward per-pair scrutiny].
- **Reading across rows.** Base/CoT/Dr.ECI all sit at recall ≥ 75 with precision ≤ 31 ; SERE trades recall for precision but stays below P=50 everywhere; SCOTECI is the only method in the balanced regime on two of three datasets.
- [TODO: add significance / resampling variance statement once final runs land.]

6 Analysis

6.1 Ablations

- Remove synthetic CoT (demos = document + gold JSON only): isolates the contribution of *reasoning-bearing* demonstrations [TODO: expected: format retained, precision drops — run].
- Remove few-shot entirely (structured protocol alone): isolates the document-level protocol [TODO: run].
- Remove the structured protocol (direct JSON answer, demos kept): isolates protocol vs. demonstrations [TODO: run].
- Selection strategy: random vs. TF-IDF vs. mention-Jaccard vs. embedding cosine [TODO: run; SERE’s finding that naive retrieval can underperform random is the hypothesis to test at document scale].

- Number of demonstrations $k \in \{1, 3, 5\}$ [TODO: run; SERE saw degradation past $k = 2$ at pair level — doc-level demos are much longer, context dilution may bite sooner].
- 2-step vs. 3-step CoT protocol [TODO: run].
- Decontamination on/off: does leakage phrasing in demos teach the model to assert without reasoning? [TODO: run — this justifies the step].
- Coherence rules on/off (in-prompt and post-hoc) [TODO: run].

6.2 Document-Level vs. Pair-Level, Same Backbone

- Same backbone forced into pair-level inference (one pair per call, same rules) vs. SCOTECI: separates the formulation effect from the backbone effect [TODO: run — this is the key internal comparison].
- Cost accounting: LLM calls and total tokens per document — pair-level: $|P|$ calls, document re-sent each time; Dr.ECI: up to 4 agents per pair; SCOTECI: 1 call (+ $2k$ amortized CoT-generation calls, cached) [TODO: measured numbers + a cost-vs-F1 figure].

6.3 Scaling with Model Quality

- Claim to substantiate: the document-level formulation *widens* its advantage as the backbone improves, because it exposes rather than bypasses long-context global reasoning.
- Model ladder: frontier closed / large open / mid open / small open [TODO: pick 4–6 from the OpenRouter matrix; plot F1 (doc-level vs pair-level) against model tier — figure, not table].
- Expected picture: weak models over-generate at document scale (precision collapse on dense docs); strong models benefit; the crossover is the point of the paper.

6.4 Error Analysis

- False-positive taxonomy: transitive flattening ($A \rightarrow B \rightarrow C$ read as $A \rightarrow C$ when the scheme does not credit it), temporal-as-causal, reporting verbs [TODO: sample and count from traces].
- Dense documents: does precision degrade with the number of gold pairs per document? [TODO: scatter/figure].
- Direction errors: *causes* vs. *causedby* confusions as a fraction of true-pair errors [TODO: compute from logs].
- Qualitative: one worked example — synthetic CoT demo, model reasoning, prediction vs. gold [TODO: pick a CTB doc].

7 Conclusion

- Reframed LLM-based ECI from per-pair scaffolding to a single document-level call with structured reasoning and coherence rules.
- Introduced synthetic, decontaminated chain-of-thought demonstrations: rationalization at inference time, no human rationales, no knowledge base, no fine-tuning.

- Large precision-driven gains where hallucination is costliest (CTB: +31 F1); balanced P/R elsewhere.
- The method is a bet on model quality: as backbones improve, the same prompt gets better — the scaffold no longer caps the model.
- Future work: full-document ESC/MAVEN runs, multilingual MECI, per-document strategy routing, teacher-generated CoT for small models.

Limitations

- ESC and MAVEN-ERE results are currently intra-sentence only; the headline CTB comparison is clean, the other two columns are not yet.
- Baseline numbers are reproduced from Hao et al. [2026], not re-run with our backbone; the same-backbone pair-level control (§6.2) is the mitigation.
- Synthetic CoT costs 2–3 generation calls per demonstration document (amortized by caching, but nonzero).
- Decontamination is itself LLM-based; residual leakage phrasing cannot be ruled out [TODO: manual audit of n sampled demos].
- Long documents with large pair sets stress the single-call format (output-length limits, attention dilution); per-document routing or chunking may be needed at the extreme.

References

- Ruichu Cai, Shengyin Yu, Jiahao Zhang, Wei Chen, Boyan Xu, and Keli Zhang. Dr.ECI: Infusing large language models with causal knowledge for decomposed reasoning in event causality identification. In *Proceedings of the 31st International Conference on Computational Linguistics (COLING 2025)*, 2025.
- Tommaso Caselli and Piek Vossen. The event StoryLine corpus: A new benchmark for causal and temporal relation extraction. In *Proceedings of the Events and Stories in the News Workshop*, 2017.
- Meiqi Chen, Yixin Cao, Kunquan Deng, Mukai Li, Kun Wang, Jing Shao, and Yan Zhang. ERGO: Event relational graph transformer for document-level event causality identification. In *Proceedings of the 29th International Conference on Computational Linguistics (COLING 2022)*, 2022.
- Qing Cheng, Zefan Zeng, Xingchen Hu, Yuehang Si, and Zhong Liu. A survey of event causality identification: Taxonomy, challenges, assessment, and prospects. *arXiv preprint*, 2025. TODO: verify final venue (ACM-style preprint).
- Jinglong Gao, Xiao Ding, Bing Qin, and Ting Liu. Is ChatGPT a good causal reasoner? A comprehensive evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.

- Zhifeng Hao, Zhongjie Chen, Junhao Lu, Shengyin Yu, Guimin Hu, Keli Zhang, Ruichu Cai, and Boyan Xu. SERE: Structural example retrieval for enhancing LLMs in event causality identification. *arXiv preprint arXiv:2605.03701*, 2026.
- Zhitao He, Pengfei Cao, Zhuoran Jin, Yubo Chen, Kang Liu, Zhiqiang Zhang, Mengshu Sun, and Jun Zhao. Zero-shot cross-lingual document-level event causality identification with heterogeneous graph contrastive transfer learning. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*, 2024. TODO: verify venue.
- Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. Distilling step-by-step! Outperforming larger language models with less training data and smaller model sizes. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2023.
- Ruijuan Hu, Jian Li, Haiyan Liu, Guilin Qi, and Yuxin Zhang. Knowledge interaction graph guided prompting for event causality identification. *Applied Intelligence*, 55:159, 2025.
- Viet Dac Lai, Amir Pouran Ben Veyseh, Minh Van Nguyen, Franck Dernoncourt, and Thien Huu Nguyen. MECI: A multilingual dataset for event causality identification. In *Proceedings of the 29th International Conference on Computational Linguistics (COLING 2022)*, 2022.
- Guixiang Liao, Yanli Chen, Wei Ke, Hanzhou Wu, and Zhicheng Dong. A contrastive representation learning framework for event causality identification. *TODO: verify journal (MDPI-style article)*, 2026.
- Cheng Liu, Wei Xiang, and Bang Wang. Identifying while learning for document event causality identification. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. TODO: verify venue (arXiv May 2024).
- Jiachang Liu, Dinghan Shen, Yizhe Zhang, Bill Dolan, Lawrence Carin, and Weizhu Chen. What makes good in-context examples for GPT-3? In *Proceedings of Deep Learning Inside Out (DeeLIO 2022): The 3rd Workshop on Knowledge Extraction and Integration for Deep Learning Architectures*, 2022.
- Sewon Min, Xinxu Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. Rethinking the role of demonstrations: What makes in-context learning work? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.
- Paramita Mirza, Rachele Sprugnoli, Sara Tonelli, and Manuela Speranza. Annotating causality in the TempEval-3 corpus. In *Proceedings of the EACL 2014 Workshop on Computational Approaches to Causality in Language (CatoCL)*, 2014.
- Lin Mu, Jun Shen, Li Ni, Lei Sang, Zhize Wu, Peiquan Jin, and Yiwen Zhang. DICP: Deep in-context prompt for event causality identification. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 2025.
- Ya Su, Hu Zhang, Guangjun Zhang, Yujie Wang, Yue Fan, Ru Li, and Yuanlong Wang. Enhancing event causality identification with LLM knowledge and concept-level event relations. In *TODO: verify venue*, 2025.

- Haoyu Wang, Fengze Liu, Jiayao Zhang, Dan Roth, and Kyle Richardson. Event causality identification with synthetic control. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024a.
- Xiaozhi Wang, Yulin Chen, Ning Ding, Hao Peng, Zimu Wang, Yankai Lin, Xu Han, Lei Hou, Juanzi Li, Zhiyuan Liu, Peng Li, and Jie Zhou. MAVEN-ERE: A unified large-scale dataset for event coreference, temporal, causal, and subevent relation extraction. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.
- Zimu Wang, Lei Xia, Wei Wang, and Xinya Du. Document-level causal relation extraction with knowledge-guided binary question answering. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024b.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Wei Xiang, Chuanhong Zhan, and Bang Wang. DAPrompt: Deterministic assumption prompt learning for event causality identification. *arXiv preprint arXiv:2307.09813*, 2023.
- Changsen Yuan, Heyan Huang, Yixin Cao, and Yonggang Wen. Discriminative reasoning with sparse event representation for document-level event-event relation extraction. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
- Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah Goodman. STaR: Bootstrapping reasoning with reasoning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Hang Zhang, Wenjun Ke, Jianwei Zhang, Zhizhao Luo, Hewen Ma, Zhen Luan, and Peng Wang. Prompt-based event relation identification with constrained prefix ATTention mechanism. *Knowledge-Based Systems*, 2023a.
- Zhuosheng Zhang, Aston Zhang, Mu Li, and Alex Smola. Automatic chain of thought prompting in large language models. In *Proceedings of the Eleventh International Conference on Learning Representations (ICLR)*, 2023b.
- R. Zhao, W. Yang, Y. Yin, et al. Causal semantics refinement model for event causality identification via information bottleneck representation learning. *Journal of King Saud University – Computer and Information Sciences*, 2026. TODO: complete author list.